

Normalizers and Sylow Theorems

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Problem 1. For finite group G , construct an injective homo $G \rightarrow GL_{|G|}(F)$, for any field F . this shows in particular that every finite group occurs as a subgroup of $GL_n(F_p)$ for some p and n . Write down $|GL_n(F_p)|$ and show that the largest power of p that divides this is $k := 1 + 2 + \dots + n - 1$. Look at that number. Exhibit a subgroup of $GL_n(F_p)$ of order p^k (look for a set of matrices where you have to choose an element of F_p k times).

Proof. $S_{|G|} \subset GL_{|G|}(F)$ as the permutation matrices. $G \subset S_{|G|}$ because G acting on G by left multiplication. Thus $G \subset GL_{|G|}(F)$. Where $G \rightarrow GL_{|G|}(F)$ sends g to its corresponding permutation matrix.

$|GL_n(F_p)| = \prod_{i=0}^{n-1} (p^n - p^i) = \prod_{i=0}^{n-1} p^i (p^{n-i} - 1) = \prod_{i=0}^{n-1} p^i \prod_{i=0}^{n-1} (p^{n-i} - 1) = p^k \prod_{i=0}^{n-1} (p^{n-i} - 1)$ $k := 1 + \dots + n - 1$.

Matrices which are upper triangular with ones on the diagonal and anything else in the triangle above the diagonal. $\square$

Problem 2. Let H be a finite group acting on a set X , with $|X|$ not divisible by p . Let p^k be the highest power of p that divides $|H|$. Show that p^k divides the order of some stabilizer.

Proof. $|H| = p^k q$ where $p \nmid q$. X may orbits which divide p and orbits which don't. $|X| = kp + |Y|$ where k is the size of the orbits which are divisible by p divided by p and $Y \subseteq X$ is the subset of elements with orbits which don't divide p . $\exists y \in Y$, because $p \nmid |X|$.

$\forall y \in |Y|$, $p \nmid |H \cdot y|$. If $|H \cdot y| = 1$, then $p^k q = |H^y| |H \cdot y| = |H^y|$ and so $p^k \mid |H^y|$. If $|H \cdot y| \neq 1$, $|H \cdot y| \mid p^k q$ and since p is prime, $|H \cdot y| \mid q$, let $a := q/|H \cdot y|$. $p^k q = |H^y| |H \cdot y| \implies p^k a |H \cdot y| = |H^k| |H \cdot y| \implies p^k a = |H^k| \implies p^k \mid |H^k|$ $\square$

Problem 3a. Now suppose H is a subgroup of G and G has a p -sylow, P . Let H act on G/P . Explain why the stabilizer of gP is $H \cap gPg^{-1}$

Proof. Suppose G acts on G/P . $\forall gP \in G/P$, $\forall h \in G^{gP}$, $hgP = gP$. Thus $hg = gp \ni p \in P$ thus $h = gpg^{-1}$. Thus $h \in gPg^{-1}$. $\forall h \in gPg^{-1}$, $hgP = gpg^{-1}gP$ for $p \in P$, $gpP = gP = hgP$. Thus $h \in G^{gP}$. Thus $G^{gP} = gPg^{-1}$

If H acts on gP , then stabilizer of gP would be the intersection of H and G^{gP} because $H \leq g$ thus $H^{gP} = H \cap gPg^{-1}$. $\square$

Problem 3b. Show that one of these is a p -sylow subgroup of H .

Proof. Since P is a p -syllow subgroup of G , $p \nmid [G : P]$. $\exists gP \in G/P \ni p \nmid |H \cdot gP|$ because otherwise $p \mid [G : P]$. By problem 2, we know that $p^k \mid |H^{gP}|$ where k is the highest power of p which divides $|H|$. Thus $p \nmid [H : H^{gP}]$ $\square$

Problem 4a. Now show that if G is a finite group and P is a syllow p , and H is a p -subgroup (meaning $|H|$ is a power of p), that H is contained in some conjugate of P .

Proof. By problem 3, $H \cap gPg^{-1}$ is p sylow in H but since H is a p group so $|H| = p^n$ so by definition $|H \cap gPg^{-1}| = p^n$ so $H = H \cap gPg^{-1}$ so $H \subseteq gPg^{-1}$ $\square$

Problem 4b. In particular conclude that all the syllow p subgroups are conjugate.

Proof. Suppose H is p sylow it has order p^k and all P sylow subgroups of G have order p^k . Since $H \subseteq gPg^{-1}$ and since $|H| = p^k$ and $|gPg^{-1}| = p^k$. Thus $H = gPg^{-1}$. $\square$

Problem 6.4. Let H be a normal subgroup of prime order p in a finite group G . Suppose that p is the smallest prime that divides the order of G . Prove that H is in the center $Z(G)$.

Proof. Let G act on $H \setminus \{e\}$ by conjugation. That gives a homomorphism of $f : G \rightarrow S_{p-1}$. $f_*(G)$ is a subgroup of S_{p-1} and thus divides S_{p-1} but $f_*(G) \mid |G|$ and p is the smallest prime which divides the order of G thus $f_*(G) = 1, p$. If $|f_*(G)| = p$ then it can't divide S_{p-1} so $|f_*(G)| = 1$. Thus everything in G when it conjugates H , $ghg^{-1} = h$ thus everything in G commutes with H . Thus $H \in Z(G)$. $\square$

Problem 6.6. Let H be a proper subgroup of a finite group G . Prove: The group G is not the union of the conjugate subgroups of H . There is a conjugacy class C that is disjoint from H .

Proof. The number of conjugate subgroups of H in G is $[G : N(H)] = |G| / |N(H)|$. But $|N(H)| \geq |H|$, so the number of conjugate subgroups is at most $|G| / |H|$. However, all conjugate subgroups share the identity element, so the size of the union of these subgroups is at most $1 + (|H| - 1)|G| / |H| = |G| + 1 - |G| / |H|$. Since $|H|$ is a proper subgroup, this value is less than $|G|$, so this proves part (a). Part (b) is obviously equivalent to part (a). $\square$

Problem 7.3. How many elements of order 5 might be contained in a group of order 20?

Proof. $|G| = 20 = 5 \cdot 4$. All p sylow groups are conjugate. G acts on the sylow p groups by conjugation and the set of sylow p groups is transitive. So number of sylow p groups must divide 20. By 3rd sylow theorems, $n =$ number of sylow p groups $n \equiv 1 \pmod{p}$. Thus n must divide 20 and must be congruent to 1 mod 5. Thus there is only 1 sylow p group and thus only 1 subgroup with order 5. Every element except e will have order 5. Thus 4 elements have order 5. $\square$

Problem 7.4a. Prove that no simple group has order pq , where p and q are prime.

Proof. Assume the contrary, let G be a simple group with order pq where p and q are prime where $q \nmid p$. Then there exists a q sylow group P with order q . The n which number of q sylow subgroups must divide pq and be congruent to 1 mod q , then n must divide p thus $n = 1$ or p . But since $q > p$, if $n = p$ $p \equiv 1 \pmod{q}$ which would mean $p = 1$ but p is prime so $n = 1$. Thus the number of conjugate subgroups is 1 and thus q sylow group is normal. Thus a contradiction forms and G is no longer simple. Thus no simple group have order pq . $\square$

Problem 7.4b. Prove that no simple group has order p^2q , where p and q are prime.

Proof. There are 1, p , p^2 q sylow subgroups.

1. If there are only 1 q -sylow subgroup, then it is normal.
2. If there are p q -sylow subgroups. Then $p \equiv 1 \pmod{q} \implies p > q$. Let n be the number of p^2 -sylow subgroups. $n \pmod{p} = 1$ and $n|p^2q$, so $n = 1, q$ since q is prime. If $n = q$ then $q \pmod{p} = 1$ which is impossible since $p > q$ and q is prime. Thus $n = 1$. Thus the p^2 - sylow subgroup is normal.
3. All the q sylow subgroups are cyclic and thus don't share any element except the identity. Thus the number of elements with order q is $p^2(q-1)$. Any element in p^2 sylow subgroup doesn't have order q . The number of elements in G without order q is $p^2q - p^2(q-1) = p^2$. Thus there is only 1 p^2 sylow subgroup which is normal.

□

Problem 7.5a. Find Sylow 2-subgroups for D_{10}

Proof. $D_{10} = \langle x, y | x^{10} = y^2 = e, yxy^{-1} = x^{-1} \rangle$ One sylow 2-subgroup is $\{e, x^5, y, x^5y\}$. All conjugations are of the form $\{e, x^5, x^k y, x^{k+5} y\}$ □

Problem 7.5b. Find Sylow 2-subgroups for T

Proof. Identity and the 3 180 degree rotations. This is Klein 4 subgroup which is normal and thus is the only 2-sylow □

Problem 7.5c. Find Sylow 2-subgroups for O

Proof. The group O has order $24 = 8 * 3 = 2^3 * 3$, so we seek a subgroup of order 8. $O \cong C_4 \times C_3$. There are 3 D_4 subgroups in O because D_4 is determined by the 4 elements in the set with a adjacency relation and there are 3 possible adjacency relation. Because for any point there is 3 elements in the set which it is not adjacent to and this fixes the rest. Thus there are 3 2 sylow subgroups each which are D_4 . □

Problem 7.5c. Find Sylow 2-subgroups for I

Proof. The group I has order 60, so we seek a subgroup of order 4. The Klein 4 subgroup in $I \cong A_5$ is a sylow subgroup. Furthermore there is only 1 as this is only 4 element subgroup of A_5 . □

Problem 7.5.6.

Proof. According to Proposition 7.7.7(c), such a group has an element x of order 7 and an element y of order 3, and these elements satisfy $xyx^{-1} = x^2$. Now the only elements of S_7 that have order 7 are the 7-cycles, so, without loss of generality, let $x = (1\ 2\ 3\ 4\ 5\ 6\ 7)$, so that $x^2 = (1\ 3\ 5\ 7\ 2\ 4\ 6)$. We must find y such that y relabels x to x^2 and has order 3. The obvious relabelling from the way x and x^2 are written, above, is $y = (2\ 3\ 5)(4\ 7\ 6)$, which indeed has order 3. Since we have the rule $yx = x^2y$, the elements $x^i y^j$ with i from 0 to 6 and j from 0 to 2 form a subgroup of S_7 of order at most 21. This subgroup contains both $\langle x \rangle$ and $\langle y \rangle$ as subgroups, so its order is divisible by 21. Therefore its order is exactly 21. □